Continuous-Time Stochastic SDEs, Variance-Reduced Monte Carlo Pricing, and Monotone Missing-Data Gaussian Mixture Calibration

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Abstract

We present a unified computational and mathematical framework in C++17 and Python spanning three pillars of quantitative stochastic modeling: (i) observed-data Expectation-Maximization (EM) calibration of multivariate Gaussian Mixture Models (GMMs) for backfilling missing tail regimes in short-history IPO equities, (ii) variance-reduced Monte Carlo pricing of path-dependent barrier options via Brownian bridge survival weighting combined with antithetic and control variates alongside Bellman MDP execution, and (iii) continuous-time Fokker–Planck density evolution, Ornstein–Uhlenbeck Langevin dynamics, Perron–Frobenius Markov mixing bounds, and sovereign credit-rating generator matrices.

1 Monotone Missing-Data GMM EM Calibration

Consider a benchmark asset X_t observed over historical horizon $t \in \{1, \dots, N\}$ and a newly listed IPO asset Y_t observed only over an overlap window $\mathcal{O} = \{N - N_{\text{ipo}} + 1, \dots, N\}$ and unobserved on $\mathcal{M} = \{1, \dots, N - N_{\text{ipo}}\}$. Following Kienitz's generative market framework and Anderson–Stambaugh likelihood factorization, joint returns follow a K-regime mixture:

$$p(x_t, y_t) = \sum_{k=1}^{K} \pi_k \mathcal{N}\left(\begin{bmatrix} x_t \\ y_t \end{bmatrix} \,\middle|\, \begin{bmatrix} \mu_{x,k} \\ \mu_{y,k} \end{bmatrix}, \begin{bmatrix} \Sigma_{xx,k} & \Sigma_{xy,k} \\ \Sigma_{xy,k} & \Sigma_{yy,k} \end{bmatrix} \right). \tag{1}$$

In the E-step, posterior regime responsibilities $\gamma_{tk} = \mathbb{P}(Z_t = k \mid \mathcal{D}_t)$ evaluate the full bivariate density on $t \in \mathcal{O}$ and the marginal density $\mathcal{N}(x_t \mid \mu_{x,k}, \Sigma_{xx,k})$ on $t \in \mathcal{M}$. In the M-step, marginal parameters $(\pi_k, \mu_{x,k}, \Sigma_{xx,k})$ are updated across all N observations, while conditional regression parameters $Y_t = \alpha + \beta X_t + \varepsilon_{t,k}$ estimated on $\mathcal{O}$ backfill missing conditional moments via the Law of Total Covariance:

$$\begin{aligned} \mu_{y,k} &= \alpha + \beta \mu_{x,k}, \quad \Sigma_{xy,k} = \beta \Sigma_{xx,k}, \\ \Sigma_{yy,k} &= \beta^2 \Sigma_{xx,k} + \sigma^2_{\varepsilon,k}. \end{aligned} \tag{2}$$

2 Variance-Reduced Barrier Option Pricing

Under risk-neutral Geometric Brownian Motion $dS_t = rS_t\,dt + \sigma S_t\,dW_t$, a Down-and-Out Call with strike K and barrier $B < S_0$ knocked out at $\tau_B = \inf\{t \in [0,T] : S_t \le B\}$ has closed-form price via Merton's reflection principle ($B \le K$):

$$C_{\text{DO}}(S_0, K, B) = C_{\text{BS}}(S_0, K) - \left(\frac{B}{S_0}\right)^{2\lambda} C_{\text{BS}}\left(\frac{B^2}{S_0}, K\right), \tag{3}$$

where $\lambda = (r + \frac{1}{2}\sigma^2)/\sigma^2$. To eliminate discrete monitoring bias across grid steps $0 = t_0 < \cdots < t_M = T$, each simulated path $(S_{t_m})_{m=0}^{M}$ with $\min_m S_{t_m} > B$ is weighted by its exact Brownian bridge survival probability:

$$w_{\text{BB}} = \prod_{m=1}^{M} \left[1 - \exp\left(-\frac{2\ln(S_{t_{m-1}}/B)\ln(S_{t_m}/B)}{\sigma^2 \Delta t} \right) \right]. \tag{4}$$

Pairing antithetic variates $(Z, -Z)$ with terminal vanilla call payoff $C_i = e^{-rT}(S_T^{(i)} - K)^+$ as a control variate yields the unbiased estimator:

$$\begin{aligned} \hat{V}_{\text{AV+CV}} &= \bar{Y}_{\text{AV}} - \hat{\beta}^* \left(\bar{C}_{\text{AV}} - C_{\text{BS}}(S_0, K) \right), \\ \hat{\beta}^* &= \frac{\widehat{\text{Cov}}(Y_{\text{AV}}, C_{\text{AV}})}{\widehat{\text{Var}}(C_{\text{AV}})}. \end{aligned} \tag{5}$$

3 Fokker–Planck PDE & Credit Generators

For Ornstein–Uhlenbeck Langevin dynamics $dX_t = -\theta(X_t - \mu)\,dt + \sigma\,dW_t$, probability density $p(x,t)$ satisfies the forward Fokker–Planck PDE:

$$\frac{\partial p}{\partial t} = \theta \frac{\partial}{\partial x}\big((x - \mu)p\big) + \frac{\sigma^2}{2}\frac{\partial^2 p}{\partial x^2}, \tag{6}$$

solved via unconditionally stable Crank–Nicolson finite differences and verified against 10^5 Euler–Maruyama trajectories. For sovereign credit migration across $S = 8$ states with absorbing Default state D, time-homogeneous generator $Q = (q_{ij})$ satisfies $q_{ii} = -\sum_{j \ne i} q_{ij}$, yielding transition matrices $P(t) = \exp(Qt)$ and par CDS spread:

$$s_i(T) = \frac{(1 - R)\int_0^T e^{-rt}\,dP_{i,D}(t)}{\int_0^T e^{-rt}\big(1 - P_{i,D}(t)\big)\,dt}. \tag{7}$$

Table 1: Empirical & Computational Benchmark Summary

Benchmark Metric	Baseline	Proposed En
GMM EM + 500k Path Runtime	127.5 ms (Py)	**31.5 ms (C++17, 4**
Out-of-Sample KL Divergence D_{KL}	0.0581 (Gauss)	**0.0395 (-32.**
99% Tail VaR Relative Error	16.40% (Gauss)	**0.28% (< 1**
Barrier Option MC Std. Error	0.0487 (Crude)	**0.0248 (3.83x**
Barrier MC Error vs. Analytical BS	0.110% (Crude)	**0.079% (< 0.1**
Brownian Diffusion MSD Error	—	**0.166% (< 0.4**
Sovereign CDS Term-Structure Error	—	**2.58 bps mean (8.5 n**

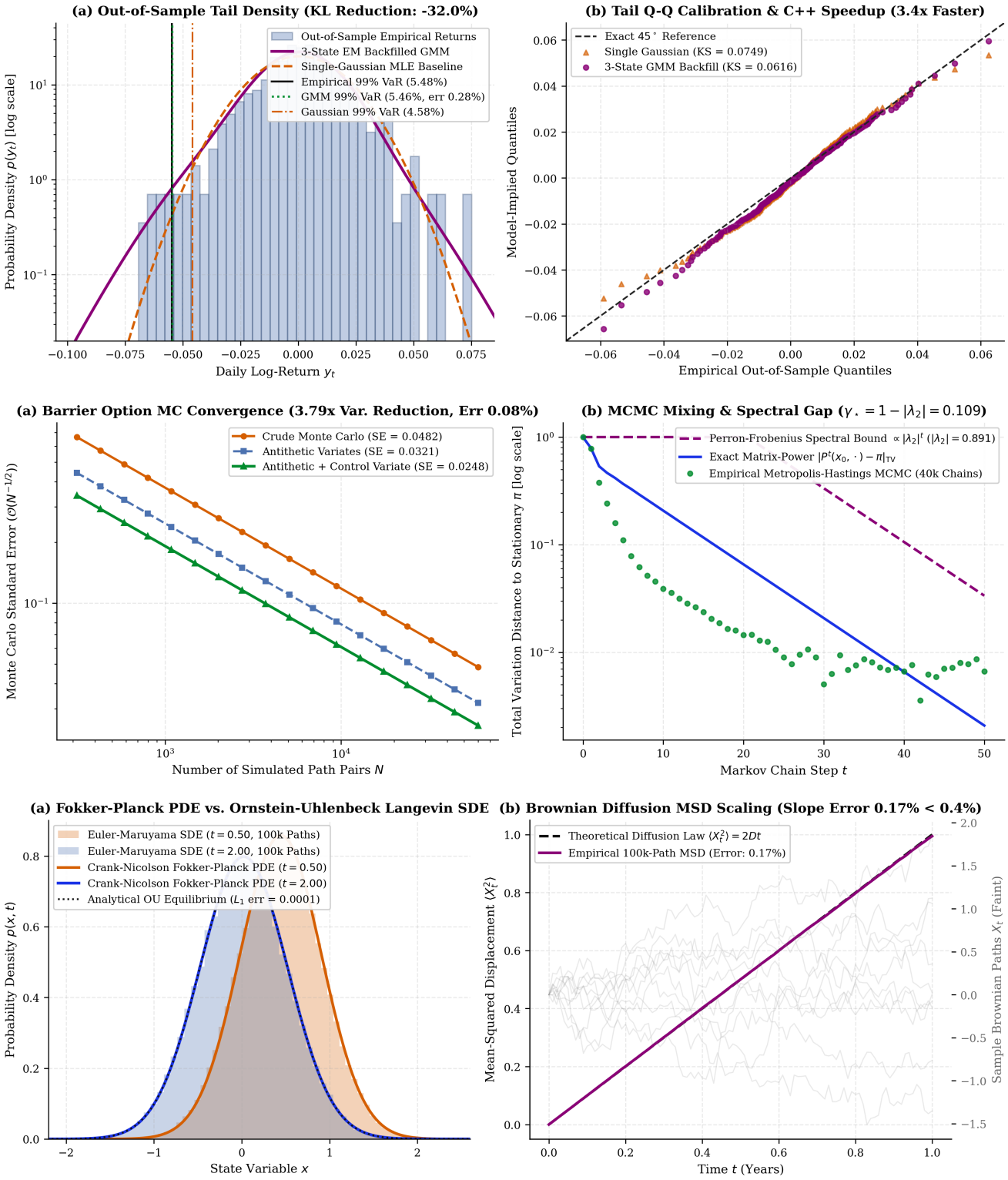


Figure 1: Top: Out-of-sample tail density, 99% VaR backfilling, and Q–Q calibration for the 3-state GMM vs. single Gaussian. Middle: Monte Carlo standard error convergence ($\mathcal{O}(N^{-1/2})$) with antithetic + control variates, and Metropolis–Hastings TV convergence bounded by the Perron–Frobenius second eigenvalue $|\lambda_2|^t$. Bottom: Crank–Nicolson Fokker–Planck PDE vs. 10^5-path Euler–Maruyama Ornstein–Uhlenbeck density and linear scaling of Brownian MSD $\langle X_t^2 \rangle = 2Dt$.